

Test 03 – Data Preprocessing Assignment (No Model Training)

Objective

The objective of this assignment is to focus only on data cleaning and preprocessing steps. No machine learning models should be trained in this part. The focus is on preparing high-quality data.

Required Data Preprocessing Steps

- 1 Understanding the dataset (shape, columns, target variable)
- 2 Checking and handling missing values
- 3 Removing duplicate records
- 4 Detecting and handling outliers
- 5 Handling incorrect data types
- 6 Handling categorical variables
- 7 Feature scaling
- 8 Removing irrelevant or redundant features
- 9 Handling skewness if present

Categorical Encoding (Recommended Methods)

Choose the best encoding method based on the nature of the data:

- 1 One-Hot Encoding – Best for nominal categories with no order (e.g., gender, color)
- 2 Ordinal Encoding – Best when categories have natural order (e.g., Low < Medium < High)
- 3 Label Encoding – Suitable only for binary categorical variables

Note: Do NOT use Label Encoding for nominal variables with more than two categories because it introduces false ordering.

Feature Scaling (Recommended Methods)

- 1 Standardization (StandardScaler) – Best for normally distributed data and distance-based models
- 2 Normalization (MinMaxScaler) – Best when values must be scaled between 0 and 1

Recommendation: StandardScaler is generally preferred for models sensitive to distance such as KNN, SVM, and Regression models.

Expected Output

A cleaned dataset ready for modeling along with clear explanation of each preprocessing step.